In [1]:
```
import numpy as num
import pandas as pd
import os
```

In [2]:
```
movies=pd.read_csv('tmdb_5000_movies.csv')
credits=pd.read_csv('tmdb_5000_credits.csv')
```

In [3]:
```
movies.head(2)
```

Out[3]:

	budget	genres	homepage	id	keywords	original_
0	237000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	[{"id": 1463, "name": "culture clash"}, {"id":...	
1	300000000	[{"id": 12, "name": "Adventure"}, {"id": 14, "...	http://disney.go.com/disneypictures/pirates/	285	[{"id": 270, "name": "ocean"}, {"id": 726, "na...	

In [4]:
```
movies.shape
```

Out[4]:
```
(4803, 20)
```

In [5]:
```
credits.head()
```

Out[5]:

	movie_id	title	cast	crew
0	19995	Avatar	[{"cast_id": 242, "character": "Jake Sully", "...	[{"credit_id": "52fe48009251416c750aca23", "de...
1	285	Pirates of the Caribbean: At World's End	[{"cast_id": 4, "character": "Captain Jack Spa...	[{"credit_id": "52fe4232c3a36847f800b579", "de...
2	206647	Spectre	[{"cast_id": 1, "character": "James Bond", "cr...	[{"credit_id": "54805967c3a36829b5002c41", "de...
3	49026	The Dark Knight Rises	[{"cast_id": 2, "character": "Bruce Wayne / Ba...	[{"credit_id": "52fe4781c3a36847f81398c3", "de...
4	49529	John Carter	[{"cast_id": 5, "character": "John Carter", "c...	[{"credit_id": "52fe479ac3a36847f813eaa3", "de...

In [6]:
```
movies = movies.merge(credits,on='title')
```

In [7]:
```
movies.head()
# budget
# homepage
# id
# original_language
# original_title
# popularity
# production_comapny
# production_countries
# release-date(not sure)
```

Out[7]:

	budget	genres	homepage	id	keywords	origin
0	237000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	[{"id": 1463, "name": "culture clash"}, {"id":...	
1	300000000	[{"id": 12, "name": "Adventure"}, {"id": 14, "...	http://disney.go.com/disneypictures/pirates/	285	[{"id": 270, "name": "ocean"}, {"id": 726, "na...	
2	245000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.sonypictures.com/movies/spectre/	206647	[{"id": 470, "name": "spy"}, {"id": 818, "name...	
3	250000000	[{"id": 28, "name": "Action"}, {"id": 80, "nam...	http://www.thedarkknightrises.com/	49026	[{"id": 849, "name": "dc comics"}, {"id": 853,...	
4	260000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://movies.disney.com/john-carter	49529	[{"id": 818, "name": "based on novel"}, {"id":...	

5 rows × 23 columns

In [8]:
```
movies = movies[['movie_id','title','overview','genres','keywords','cast','c
```

In [9]:
```
movies.head()
```

Out[9]:

	movie_id	title	overview	genres	keywords	cast	
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	[{"id": 1463, "name": "culture clash"}, {"id":...	[{"cast_id": 242, "character": "Jake Sully", "...	[{"cre "52fe48009251416c750
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[{"id": 12, "name": "Adventure"}, {"id": 14, "...	[{"id": 270, "name": "ocean"}, {"id": 726, "na...	[{"cast_id": 4, "character": "Captain Jack Spa...	[{"cre "52fe4232c3a36847f80
2	206647	Spectre	A cryptic message from Bond’s past sends him o...	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	[{"id": 470, "name": "spy"}, {"id": 818, "name...	[{"cast_id": 1, "character": "James Bond", "cr...	[{"cre "54805967c3a36829b50
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...	[{"id": 28, "name": "Action"}, {"id": 80, "nam...	[{"id": 849, "name": "dc comics"}, {"id": 853,...	[{"cast_id": 2, "character": "Bruce Wayne / Ba...	[{"cre "52fe4781c3a36847f81
4	49529	John Carter	John Carter is a war-weary, former military ca...	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	[{"id": 818, "name": "based on novel"}, {"id":...	[{"cast_id": 5, "character": "John Carter", "c...	[{"cre "52fe479ac3a36847f81

In [10]:
```
import ast
```

In [11]:
```
def convert(text):
    L = []
    for i in ast.literal_eval(text):
        L.append(i['name'])
    return L
```

In [12]:
```
movies.dropna(inplace=True)
```

In [13]:
```
movies['genres'] = movies['genres'].apply(convert)
movies.head()
```

Out[13]:

	movie_id	title	overview	genres	keywords	cast	
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[{"id": 1463, "name": "culture clash"}, {"id":...	[{"cast_id": 242, "character": "Jake Sully", "...	[{"cred "52fe48009251416c750ac
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[{"id": 270, "name": "ocean"}, {"id": 726, "na...	[{"cast_id": 4, "character": "Captain Jack Spa...	[{"cred "52fe4232c3a36847f800b
2	206647	Spectre	A cryptic message from Bond's past sends him o...	[Action, Adventure, Crime]	[{"id": 470, "name": "spy"}, {"id": 818, "name...	[{"cast_id": 1, "character": "James Bond", "cr...	[{"cred "54805967c3a36829b5002
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...	[Action, Crime, Drama, Thriller]	[{"id": 849, "name": "dc comics"}, {"id": 853,...	[{"cast_id": 2, "character": "Bruce Wayne / Ba...	[{"cred "52fe4781c3a36847f8139
4	49529	John Carter	John Carter is a war-weary, former military ca...	[Action, Adventure, Science Fiction]	[{"id": 818, "name": "based on novel"}, {"id":...	[{"cast_id": 5, "character": "John Carter", "c...	[{"cred "52fe479ac3a36847f813e

In [14]:
```
movies['keywords'] = movies['keywords'].apply(convert)
movies.head()
```

Out[14]:

	movie_id	title	overview	genres	keywords	cast	
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[{"cast_id": 242, "character": "Jake Sully", "...	[{"cred "52fe48009251416c750ac
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drug abuse, exotic island, east india ...	[{"cast_id": 4, "character": "Captain Jack Spa...	[{"cred "52fe4232c3a36847f800b
2	206647	Spectre	A cryptic message from Bond's past sends him o...	[Action, Adventure, Crime]	[spy, based on novel, secret agent, sequel, mi...	[{"cast_id": 1, "character": "James Bond", "cr...	[{"cred "54805967c3a36829b5002
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...	[Action, Crime, Drama, Thriller]	[dc comics, crime fighter, terrorist, secret i...	[{"cast_id": 2, "character": "Bruce Wayne / Ba...	[{"cred "52fe4781c3a36847f8139
4	49529	John Carter	John Carter is a war-weary, former military ca...	[Action, Adventure, Science Fiction]	[based on novel, mars, medallion, space travel...	[{"cast_id": 5, "character": "John Carter", "c...	[{"cred "52fe479ac3a36847f813e

In [17]:
```
import ast
ast.literal_eval('[{"id": 28, "name": "Action"}, {"id": 12, "name": "Adventu
```

Out[17]:
```
[{'id': 28, 'name': 'Action'},
 {'id': 12, 'name': 'Adventure'},
 {'id': 14, 'name': 'Fantasy'},
 {'id': 878, 'name': 'Science Fiction'}]
```

In [18]:
```
def convert3(text):
    L = []
    counter = 0
    for i in ast.literal_eval(text):
        if counter < 3:
            L.append(i['name'])
        counter+=1
    return L
```

In [19]:
```
movies['cast'] = movies['cast'].apply(convert)
movies.head()
```

Out[19]:

	movie_id	title	overview	genres	keywords	cast	
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[Sam Worthington, Zoe Saldana, Sigourney Weave...	[{"cre "52fe48009251416c750a
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drug abuse, exotic island, east india ...	[Johnny Depp, Orlando Bloom, Keira Knightley, ...	[{"cre "52fe4232c3a36847f80(
2	206647	Spectre	A cryptic message from Bond's past sends him o...	[Action, Adventure, Crime]	[spy, based on novel, secret agent, sequel, mi...	[Daniel Craig, Christoph Waltz, Léa Seydoux, R...	[{"cre "54805967c3a36829b50(
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...	[Action, Crime, Drama, Thriller]	[dc comics, crime fighter, terrorist, secret i...	[Christian Bale, Michael Caine, Gary Oldman, A...	[{"cre "52fe4781c3a36847f813
4	49529	John Carter	John Carter is a war-weary, former military ca...	[Action, Adventure, Science Fiction]	[based on novel, mars, medallion, space travel...	[Taylor Kitsch, Lynn Collins, Samantha Morton,...	[{"cre "52fe479ac3a36847f813

In [20]:
```
movies['cast'] = movies['cast'].apply(lambda x:x[0:3])
```

In [21]:
```
def fetch_director(text):
    L = []
    for i in ast.literal_eval(text):
        if i['job'] == 'Director':
            L.append(i['name'])
    return L
```

In [22]:
```
movies['crew'] = movies['crew'].apply(fetch_director)
```

In [23]:
```
#movies['overview'] = movies['overview'].apply(lambda x:x.split())
movies.sample(5)
```

Out[23]:

	movie_id	title	overview	genres	keywords	cast	crew
4449	15708	Latter Days	Aaron Davis (Steve Sandvoss) and Christian Mar...	[Drama, Comedy, Romance]	[gay, coming out, religion, mormon]	[Steve Sandvoss, Wes Ramsey, Jacqueline Bisset]	[C. Jay Cox]
1723	112949	Safe Haven	A young woman with a mysterious past lands in ...	[Romance]	[based on novel, small town, widower, single f...	[Julianne Hough, Josh Duhamel, Cobie Smulders]	[Lasse Hallström]
2763	1251	Letters from Iwo Jima	The story of the battle of Iwo Jima between th...	[Action, Adventure, Drama, War]	[world war ii, cave, dying and death, pacific ...	[Ken Watanabe, Kazunari Ninomiya, Tsuyoshi Ihara]	[Clint Eastwood]
2336	10625	Mean Girls	Cady Heron is a hit with The Plastics, the A-l...	[Comedy]	[female friendship, high school, fish out of w...	[Lindsay Lohan, Rachel McAdams, Tim Meadows]	[Mark Waters]
3429	34723	The Adventures of Huck Finn	Climb aboard for an extraordinary version of M...	[Action, Adventure, Drama, Family]	[southern accent, wanted man, lynch mob, refle...	[Elijah Wood, Courtney B. Vance, Robbie Coltrane]	[Stephen Sommers]

In [24]:
```
def collapse(L):
    L1 = []
    for i in L:
        L1.append(i.replace(" ",""))
    return L1
```

In [25]:
```
movies['cast'] = movies['cast'].apply(collapse)
movies['crew'] = movies['crew'].apply(collapse)
movies['genres'] = movies['genres'].apply(collapse)
movies['keywords'] = movies['keywords'].apply(collapse)
```

In [26]:
```
movies.head()
```

Out[26]:

	movie_id	title	overview	genres	keywords	cast	
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, ScienceFiction]	[cultureclash, future, spacewar, spacecolony, ...	[SamWorthington, ZoeSaldana, SigourneyWeaver]	[JamesC
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drugabuse, exoticisland, eastindiatrad...	[JohnnyDepp, OrlandoBloom, KeiraKnightley]	[GoreV
2	206647	Spectre	A cryptic message from Bond’s past sends him o...	[Action, Adventure, Crime]	[spy, basedonnovel, secretagent, sequel, mi6, ...	[DanielCraig, ChristophWaltz, LéaSeydoux]	[Sam
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...	[Action, Crime, Drama, Thriller]	[dccomics, crimefighter, terrorist, secretiden...	[ChristianBale, MichaelCaine, GaryOldman]	[Christophe
4	49529	John Carter	John Carter is a war-weary, former military ca...	[Action, Adventure, ScienceFiction]	[basedonnovel, mars, medallion, spacetravel, p...	[TaylorKitsch, LynnCollins, SamanthaMorton]	[Andrew

In [27]:
```
movies['overview'] = movies['overview'].apply(lambda x:x.split())
```

In [28]:
```
movies['tags'] = movies['overview'] + movies['genres'] + movies['keywords']
```

In [29]:
```
new = movies.drop(columns=['overview','genres','keywords','cast','crew'])
#new.head()
```

In [30]:
```
new['tags'] = new['tags'].apply(lambda x: " ".join(x))
new.head()
```

Out[30]:

	movie_id	title	tags
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...
2	206647	Spectre	A cryptic message from Bond’s past sends him o...
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...
4	49529	John Carter	John Carter is a war-weary, former military ca...

In [31]:
```
from sklearn.feature_extraction.text import CountVectorizer
cv = CountVectorizer(max_features=5000,stop_words='english')
```

In [32]:
```
vector = cv.fit_transform(new['tags']).toarray()
```

In [33]:
```
vector.shape
```

Out[33]:
```
(4806, 5000)
```

In [34]:
```
from sklearn.metrics.pairwise import cosine_similarity
```

In [35]:
```
similarity = cosine_similarity(vector)
```

In [36]:
```
similarity
```

Out[36]:
```
array([[1.        , 0.08964215, 0.06071767, ..., 0.02519763, 0.0277885 ,
        0.        ],
       [0.08964215, 1.        , 0.06350006, ..., 0.02635231, 0.        ,
        0.        ],
       [0.06071767, 0.06350006, 1.        , ..., 0.02677398, 0.        ,
        0.        ],
       ...,
       [0.02519763, 0.02635231, 0.02677398, ..., 1.        , 0.07352146,
        0.04774099],
       [0.0277885 , 0.        , 0.        , ..., 0.07352146, 1.        ,
        0.05264981],
       [0.        , 0.        , 0.        , ..., 0.04774099, 0.05264981,
        1.        ]])
```

In [37]:
```
new[new['title'] == 'The Lego Movie'].index[0]
```

Out[37]:
```
744
```

In [38]:
```
def recommend(movie):
    index = new[new['title'] == movie].index[0]
    distances = sorted(list(enumerate(similarity[index])),reverse=True,key =
    for i in distances[1:6]:
        print(new.iloc[i[0]].title)
```

In [40]:
```
recommend('Batman')
```

```
Batman
Batman & Robin
The Dark Knight Rises
Batman Begins
Batman Returns
```

In [41]:
```
import pickle
```

In [42]:
```
pickle.dump(new,open('movie_list.pkl','wb'))
pickle.dump(similarity,open('similarity.pkl','wb'))
```